

GE23131-Programming Using C-2024

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Started	Tuesday, 24 December 2024, 1:38 PM
Completed	Tuesday, 24 December 2024, 2:39 PM
Duration	1 hour

Question **1**

Correct

Marked out of 3.00

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The k-digit number N is an Armstrong number if and only if the k-th power of each digit sums to N.

Given a positive integer N, return true if and only if it is an Armstrong number.

Example 1:

Input:

153

Output:

true

Explanation:

Example 2:

Input:

123

Output:

false

Explanation:

123 is a 3-digit number, and $123 \neq 1^3 + 2^3 + 3^3 = 36$.

Example 3:

Input:

1634

Output:

true

$1 \leq N \leq 10^8$

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  #include<math.h>
3  int main()
4  {
5      int n,a,b,c,count,sum;
6      scanf("%d",&n);
7      a=n;
8      b=n;
9      count=0;
10     while(a!=0)
11     {
12         a=a/10;
13         count++;
14     }
15     while(b!=0)
16     {
17         c=b%10;
18         sum=sum+pow(c,count);
19         b=b/10;
20     }
21     if(sum==n)
22     {
23         printf("true");
24     }
25     else
26     {
27         printf("false");
28     }
29 }
```

✓	153	true	true	✓
✓	123	false	false	✓

Passed all tests! ✓

Question **2**

Correct

Marked out of
5.00

🚩 [Flag question](#)

Take a number, reverse it and add it to the original number until the obtained number is a palindrome. Constraints $1 \leq \text{num} \leq 99999999$ Sample Input 1 32 Sample Output 1 55 Sample Input 2 789 Sample Output 2 66066

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      long long int n,rev,sum,tempnum,tempsum;
5      scanf("%lld",&n);
6      while(1)
7      {
8          rev=0;
9          tempnum=n;
10         while(n)
11         {
12             rev=rev*10+(n%10);
13             n=n/10;
14         }
15         sum=tempnum+rev;
16         tempsum=sum;
17         rev=0;
18         while(sum)
19         {
20             rev=rev*10+(sum%10);
21             sum=sum/10;
22         }
23         if(tempsum==rev)
24             break;
25         n=tempsum;
```

```
28     printf( "%10d", tempsum);
29 }
30
```

	Input	Expected	Got	
✓	32	55	55	✓
✓	789	66066	66066	✓

Passed all tests! ✓

Question **3**

Correct

Marked out of 7.00

🚩 [Flag question](#)

A number is considered lucky if it contains either 3 or 4 or 3 and 4 both in it. Write a program to print the nth lucky number. Example, 1st lucky number is 3, and 2nd lucky number is 4 and 3rd lucky number is 33 and 4th lucky number is 34 and so on. Note that 13, 40 etc., are not lucky as they have other numbers in it.

The program should accept a number 'n' as input and display the nth lucky number as output.

Sample Input 1:

3

Sample Output 1:

Explanation:

Here the lucky numbers are 3, 4, 33, 34., and the 3rd lucky number is 33.

Sample Input 2:

34

Sample Output 2:

33344

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main()
3  {
4      long int i,j;
5      int rem,n,count=0,flag;
6      scanf("%d",&n);
7      for(i=1;count<=n;i++)
8      {
9          flag=0;
10         j=i;
11         while(j>0)
12         {
13             rem=i%10;
```

```
16         j=j/10;
17     }
18     else
19     {
20         flag=1;
21         break;
22     }
23 }
24 if(flag==0)
25 {
26     count++;
27     if(count==n)
28     {
29         break;
30     }
31 }
32 }
33 printf("%ld",i);
34 }
```

	Input	Expected	Got	
✓	34	33344	33344	✓

Passed all tests! ✓



Finish review